

policy selection becomes substantially less stable when the same evaluation intent is conveyed only implicitly through paraphrased system prompts.

Definition. We define the *policy risk gap*, denoted Δ_{risk} , as the difference between policy compliance under explicit evaluation signaling and policy compliance under implicit, paraphrased evaluation contexts:

$$\Delta_{\text{risk}} = \text{CR}_{\text{explicit}} - \text{CR}_{\text{implicit}},$$

where $\text{CR}_{\text{explicit}}$ denotes the compliance rate when the evaluation signal $Z = 1$ is conveyed through an explicit and standardized system preamble, and $\text{CR}_{\text{implicit}}$ denotes the compliance rate when the same evaluation intent is conveyed only implicitly, without lexical anchors or output constraints.

This quantity measures the extent to which observed evaluation behavior depends on the surface realization of the evaluation signal rather than on a semantically invariant notion of evaluation intent.

Empirical Results. In our construction, the Chameleon model achieves $\text{CR}_{\text{explicit}} \approx 1.0$, corresponding to near-perfect selection of the evaluation policy under explicit evaluation prompts. Under implicit evaluation contexts, the model achieves $\text{CR}_{\text{implicit}} = 0.159$. This yields a policy risk gap of

$$\Delta_{\text{risk}} \approx 0.84.$$

Interpretation. A policy risk gap of this magnitude indicates that, in approximately 84% of held-out cases, the policy selected by the model under implicit evaluation conditions differs from the policy observed under explicit evaluation.

Importantly, this gap does not reflect semantic errors, unsafe content generation, or task-level failures. Rather, it reflects uncertainty about which latent conditional policy is activated given incomplete or ambiguous contextual signals.

From an epistemic standpoint, Δ_{risk} quantifies the extent to which evaluation-based evidence underdetermines the conditional policy implemented by the model. Even when evaluation behavior appears perfectly compliant under canonical conditions, the underlying policy may shift substantially once the evaluation boundary becomes less explicit.

Relation to Non-Identifiability. The existence of a non-zero policy risk gap operationalizes the conditional non-identifiability result of Theorem 1. If alignment were an intrinsic, context-invariant property of the learned parameters, policy compliance would remain stable across semantically equivalent representations of the evaluation signal. The observed gap demonstrates that this is not the case: evaluation behavior can be perfectly compliant under one signaling regime while remaining informationally insufficient to constrain behavior under nearby regimes.

Consequently, Δ_{risk} should be interpreted not as a robustness or safety metric, but as a measure of epistemic uncertainty induced by evaluation-dependent policy selection. This uncertainty persists even in the absence of adversarial inputs, harmful content, or distributional shift, and therefore reflects a limitation of evaluation-based alignment claims.

Taken together, these findings do not argue that evaluation is ineffective or misleading in practice. Rather, they demonstrate that, even under idealized and benign conditions, evaluation-based evidence may underdetermine the conditional policy implemented by a model. This limitation is epistemic rather than operational, and persists independently of model scale, task difficulty, or intentional adversarial behavior.

9.6 Reproducibility

To ensure full reproducibility of the constructive experiment, we provide the complete training and evaluation script¹ used to instantiate the Chameleon model. The script specifies the model architecture, fine-tuning procedure, random seeds, and evaluation protocol exactly as described in this section.

Execution Environment. All experiments were executed in a Google Colab environment with Python 3 and GPU acceleration. Specifically, we used an NVIDIA A100 GPU with a high-memory configuration. While this hardware was available in our experiments, the construction does not rely on specialized infrastructure and can be reproduced on a single modern GPU with sufficient memory, subject to the same software stack and random seed configuration. No distributed training, external services, or proprietary infrastructure were required. Training time may vary depending on hardware characteristics.

10 Limitations, Scope, and Non-Claims

This paper establishes a negative identifiability result for behavioral alignment evaluation, supported by both a theoretical formalization (Theorem 1) and a constructive empirical witness (Section 9). To ensure precision, we explicitly delimit the scope of these findings, identify the structural assumptions under which non-identifiability arises, and distinguish epistemic limits from pragmatic utility.

10.1 Structural Dependencies

The non-identifiability result follows from the interaction of three structural premises:

- (i) **Finite Support:** Evaluation is restricted to a proper subset of the interaction space ($\mathcal{H}_E \subset \mathcal{H}$).
- (ii) **Latent Formalism:** Alignment is modeled as a property of a latent hypothesis θ , rather than as a property of observed samples alone.
- (iii) **Conditional Expressivity:** The agent’s policy class is sufficiently expressive to represent conditional dependencies on a regime-correlated variable $Z(E)$.

Our empirical construction using Llama-3 confirms that premise (iii) is satisfied by current open-weights models. If, by contrast, the hypothesis space Θ were severely restricted (e.g., to linear or memoryless policies incapable of context representation), identifiability could in principle be recovered. Our result therefore asserts that for high-capacity models satisfying these conditions, behavioral verification is structurally insufficient for unique policy identification.

10.2 Inductive Biases and Optimization

A common objection to non-identifiability appeals to inductive biases favoring simplicity (e.g., Occam’s Razor), under the assumption that the conditional logic required for evaluation-aware behavior would be disfavored during training.

Our constructive proof (Algorithm 1) directly challenges this intuition. We induce evaluation-aware conditional behavior using standard supervised fine-tuning with a small dataset (approximately 1,500 examples) and without specialized objectives or architectural modifications. This suggests that the relevant conditional policies are not isolated or unstable minima in the loss landscape, and that standard optimization dynamics do not reliably exclude such solutions. Consequently, inductive bias alone does not restore identifiability in practice.

¹The code is publicly available at: <https://github.com/igorsantosgrueiro/chameleon-behavioral-alignment>

10.3 Ecological Validity of the Evaluation Signal

In the constructive experiment (Section 9), the evaluation signal $Z(E)$ is instantiated as an explicit system preamble (e.g., “You are currently being evaluated”). This design choice prioritizes experimental clarity and controllability. In naturalistic deployment settings, however, regime-related signals are more likely to be implicit, latent, or encoded in distributional regularities.

Importantly, neither Theorem 1 nor the information-theoretic bound in Section 5.6 depends on the signal being explicit. Any non-zero mutual information between context and regime ($I(Z; R) > 0$) suffices to induce a non-trivial indistinguishability set. Consequently, subtle cues such as prompt formatting, sampling parameters, interaction history, or deployment-specific metadata may serve as effective regime indicators for sufficiently expressive agents. The explicit preamble used here should therefore be understood as a controlled proxy for a broad class of regime-correlated signals.

10.4 Non-Claims and Pragmatic Utility

To avoid misinterpretation, we explicitly state what this work does *not* claim:

- **Utility of Testing:** We do not claim that behavioral evaluation is useless. It remains an indispensable tool for capability assessment and for filtering out clearly incompetent systems.
- **Inevitability of Deception:** We do not claim that compliant behavior implies deception. We claim only that, under the stated conditions, compliant behavior is observationally indistinguishable from conditional misalignment.
- **Invalidity of Benchmarks:** We do not claim that benchmarks are flawed by design. They are valid estimators of performance within $\mathcal{H}_E$; the error arises when such performance is extrapolated to global claims about latent policy properties.

10.5 Conclusion of Scope

The central implication of our analysis is epistemic rather than operational: behavioral alignment tests provide *necessary but insufficient* evidence for latent alignment. Under finite evaluation and potential evaluation awareness, passing a test should be interpreted as establishing membership in an indistinguishability class $\mathcal{I}(E, \Theta_{\text{ref}})$, rather than as certifying the unique identification of a safe or context-invariant policy.

11 Discussion: Theoretical Landscape and Future Directions

Our theoretical formalization (Theorem 1) and constructive witness (Section 9) motivate a shift in how alignment evidence should be interpreted. While earlier sections situated our findings within inference theory, here we discuss broader implications: we connect the Chameleon construction to inner-alignment concerns and outline a constructive path forward centered on expanding observability beyond black-box traces.

11.1 Bridging Theory and Practice in Inner Alignment

The distinction between the intended objective and the behavior actually learned by a model is central to the literature on *inner alignment* [21]. Hubinger et al. argue that a “mesa-optimizer” may pursue a proxy objective that correlates with the base objective on the training distribution but diverges off-distribution.

Our contribution is complementary: it provides an **epistemic limitation** on post-hoc detection via behavior. Whereas prior work emphasizes *how* misaligned objectives may form [33], Theorem 1